

versions of Whitney's flat cochains. The next paragraph provides an alternative description of $\mathbf{F}_m^\alpha(K)^*$.

An α -fractional charge ω over K is a linear functional on $\mathbf{N}_m(K)$, the space of normal m -currents supported in K , that satisfies the continuity condition

$$|\omega(T)| \leqslant C \mathbf{N}(T)^{1-\alpha} \mathbf{F}(T)^\alpha \quad (1)$$

for all $T \in \mathbf{N}_m(K)$. When K is sufficiently nice, for instance if it is convex, then ω extends by density to a continuous linear functional on $\mathbf{F}_m^\alpha(K)$. Conversely, any element of $\mathbf{F}_m^\alpha(K)^*$, restricted to $\mathbf{N}_m(K)$, satisfies (1). Thus, the space $\mathbf{CH}^{m,\alpha}(K)$ of α -fractional m -charges is (isomorphic to) the dual of $\mathbf{F}_m^\alpha(K)$ in the case where K is convex.

One important class of α -fractional charges consists of α -Hölder continuous differential forms. More precisely, to each such form $\eta \in \text{Lip}^\alpha(K; \wedge^m \mathbb{R}^d)$, one associates the functional $\Lambda(\eta)$ that acts on normal currents $T \in \mathbf{N}_m(K)$ by

$$\Lambda(\eta)(T) = \int_K \langle \eta(x), \vec{T}(x) \rangle d\|T\|(x).$$

In the simplest case where $m = 0$, the map Λ is in fact a Banach space isomorphism between $\text{Lip}^\alpha(K)$ and $\mathbf{CH}^{0,\alpha}(K)$. As a result, we can identify α -fractional 0-charges over K with α -Hölder continuous functions on K . In this case, it is known that $\text{Lip}^\alpha(K)$ is the double dual of the little Hölder space, described for example in [28, Chapter 4]. We treat the general case $m \geqslant 0$ by analogy. We define the space of little α -fractional m -charges $\mathbf{ch}^{m,\alpha}(K)$ as the closure in $\mathbf{CH}^{m,\alpha}(K)$ of $\{\Lambda(\omega|_K) : \omega \text{ smooth}\}$, and we establish that $\mathbf{F}_m^\alpha(K) \cong \mathbf{ch}^{m,\alpha}(K)^*$. Thus any bounded sequence in $\mathbf{F}_m^\alpha(K)$ possesses a weakly convergent subsequence, by the Banach-Alaoglu theorem. Convergence in flat norm is later obtained by adapting the classical deformation theorem to the setting of fractional currents.

1.3. Generalized Stokes' formula. The exterior derivative operator for fractional charges is well-defined as the adjoint of the boundary operator. Specifically, $d\omega$ is the α -fractional $(m+1)$ -charge over K , given by

$$\langle d\omega, T \rangle = \langle \omega, \partial T \rangle \quad (2)$$

for $T \in \mathbf{N}_m(K)$. If K is convex, this formula extends more generally to α -fractional currents over K .

A particular case of this formula is worthy of attention, as it results in a generalized Gauss-Green formula for Hölder vector fields over fractal boundaries. It extends the work of Harrison and Norton [17] (it should be however stressed that formula (2) is purely definitional, contrary to the more constructive Gauss-Green formulas in [17, 16]). Indeed, we can take for U a bounded open set with finite $(1-\alpha)$ -perimeter (meaning that $\mathbb{1}_U \in W_c^{1-\alpha,1}(\mathbb{R}^d)$, or equivalently, $\llbracket U \rrbracket$ is α -fractional). We shall prove that this condition is weaker than the $(d-1+\alpha)$ -summability (see [17, §3] or Subsection 3.2) of ∂U that is required in the Gauss-Green formula of Harrison-Norton. In this case, formula (2) applies for $T = \llbracket U \rrbracket$ and ω that is associated to an α -Hölder continuous $(d-1)$ -form.

1.4. Pushforward by Hölder maps. The pushforward of currents by Hölder maps f has been explored in the literature, notably in [14] and [29]. The results in [29] complement ours, as they address the case where the components f_i of f have different Hölder exponents.

Our contribution is to provide a functional-analytical setting to this construction by showing that $f_\#$ is a bounded linear map between spaces of fractional currents. Indeed, in Section 8, we define the pushforward $f_\#T$ of an α -fractional m -current T by a γ -Hölder continuous map f . The construction makes sense as long as $m+\alpha < \gamma(m+1)$. Under this assumption, we show that $f_\#T$ is β -fractional, where β is given by the equation

$$\frac{m+\alpha}{\gamma} = m+\beta.$$